MASTER'S THESIS · 2014

Parametric Adaptive Breakout Trading Bot

Automated decision-making for stock market portfolio management using a daily volatility filter breakout strategy implemented on NYSE equities

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2011–2013 · 2.5 Years of Data

+75.9%
Net Profit
on $50K Starting Capital

24,238
Total Trades
Over 2.5 Years

2.05
Sharpe Ratio
(Excellent ≥ 2.0 by Sharpe Method)

4.28
Recovery Factor
(2× Minimum by Yoshito Method)

−13.5%
Max Drawdown
Risking 13.5% to Earn 75.9%

ABSTRACT

"Computers are now being used to generate news stories about company earnings results or economic statistics as they are released. And this almost instantaneous information forms a direct feed into other computers which trade on the news."

— Financial Times

This thesis explores how modern software solutions facilitate automated decision-making on stock markets. It proposes and implements an **Adaptive Reversal Parametric Breakout strategy** based on a Daily Volatility Filter, deployed via a C#/S# + Wealth-Lab automated trading system. The strategy was tested on 80 NYSE and MICEX stocks, filtered to a 25-stock NYSE portfolio, and backtested on 2.5 years of real market data — generating 75.9% net profit with a 13.5% maximum drawdown.

RESEARCH FRAMEWORK

Five Research Questions

RQ 01

Active or passive portfolio management?

Markets are inefficient (Haugen 1995; Rosenberg & Reid 1998). Active management exploits inefficiencies — automated systems are essential to detect them in real-time.

RQ 02

Technical or fundamental analysis?

FA suits passive "buy & hold." TA suits active trading — it is formalizable, computationally tractable, and automatable. TA is the foundation for trading robots.

RQ 03

What trading strategies exist & how to classify them?

Two broad classes: **Execution** (Iceberg, TWAP, VWAP — order splitting) and **Speculative** (Trendline, Pullback, Breakout, Arbitrage, Scalping). Breakout offers the best automation trade-off.

RQ 04

What platform & architecture for trading robots?

C# + S# (Stock Sharp) + Wealth-Lab provides multi-core HFT support, DMA connectivity, and emulation >1M events/sec. Six infrastructure options analyzed (A–F), with DMA recommended.

RQ 05

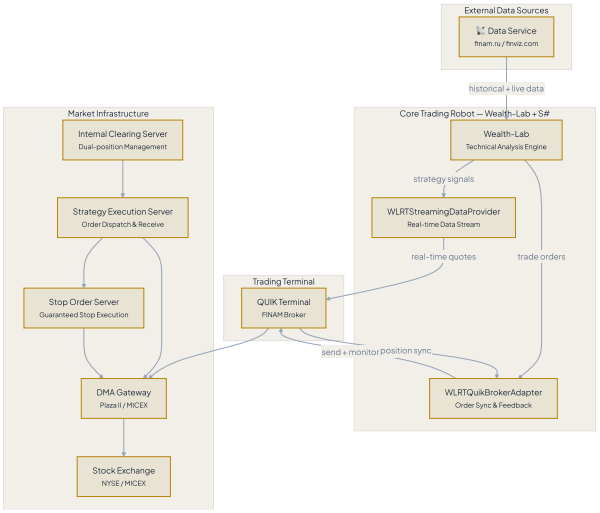
Which strategy to choose for successful automated trading?

The ARP-Breakout strategy: **simple, reversible, adaptive.** It learns the daily trading range via a volatility filter, enters on breakouts, and manages positions with trailing stops tied to ATR — all controlled through 14 optimized parameters.

SYSTEM ARCHITECTURE

Automated Trading System Infrastructure

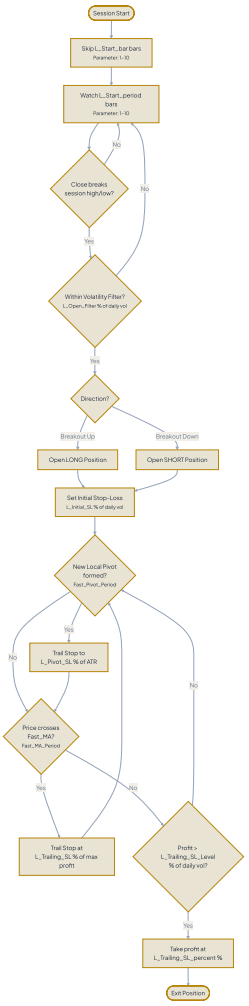
The recommended architecture for active portfolio management using direct market access (DMA) via Plaza II or MICEX gateway.



STRATEGY LOGIC

ARP-Breakout Strategy Flow

The strategy skips early bars, watches for breakouts within the daily volatility range, and manages exits with multi-level trailing stops.



STRATEGY PARAMETERS

14 Parameters for Optimization

Each parameter was optimized using the Non-Gradient Monte Carlo Method to avoid brute-force computation (which would require ~2×10^{50} iterations).

#	PARAMETER	PURPOSE	RANGE
1	`Fast Pivot Period`	Period for detecting local pivot extremes	2–7
2	`Fast MA Period`	Period for simple moving average of closing price	0–20
3	`L Start Bar`	Bars to skip after session open before monitoring begins	1–10
4	`L Start Period`	Period for checking position-open conditions	1–10
5	`L Back Period`	Check period for price change at position open	1–6
6	`L Initial SL`	Initial stop-loss as % of daily volatility (first entry)	50–300
7	`L Initial SL 2`	Initial stop-loss as % of daily vol (re-entry after loss)	50–300
8	`L Pivot SL`	Stop level after new local extremum, % of ATR from last pivot	50–200
9	`L Max Point SL`	Maximum drawdown in points relative to max profit	10–50
10	`L Trailing SL Level`	% of daily vol at which trailing stop activates	50–200
11	`L Trailing SL Percent`	% of max profit to take when trailing stop triggers	10–60
12	`L Open Filter`	Max % of daily vol the price may move to open a position	10–100
13	`L Open Filter 2`	Max % of daily vol in opposite direction to open	10
14	`L Open Filter 3`	ATR/daily vol ratio to allow pivot re-entry after loss	10–100

PORTFOLIO SELECTION

Stock Filtering: NYSE vs. MICEX

80 stocks tested across two exchanges. Only those with non-negative strategy returns were selected into the final portfolio.

Filtering Principle: Rather than selecting by P/E ratio or market cap, stocks are selected by **strategy compatibility** — the strategy itself determines which stocks belong in the portfolio. The portfolio can be rebalanced periodically by re-running the filter.

MICEX (RUSSIAN EXCHANGE)

7 / 40

Stocks passed the strategy filter

GAZP, SBER03, MRKH, URKA, GRAZ, SVAV, MNOD

Unsuitable — quality of trades too low

NYSE (NEW YORK STOCK EXCHANGE)

25 / 40

Stocks passed the strategy filter

AAPL, ADBE, AIG, AMT, BA, BRCM, CAT, CSCO, CVX, DD, DIS, FSLR, GE, GOOG, HPQ, IBM, INTC, JNJ, JPM, KO, MCD, MSFT, PG, T, VZ

Selected — 25-stock NYSE portfolio formed

BACKTEST RESULTS

Performance on Real Market Data

Portfolio simulation over 2.5 years of NYSE data (2011–2013) with $50,000 starting capital. Includes commission costs.

METRIC	ALL TRADES	LONG	SHORT
Starting Capital	$50,000	$50,000	$50,000
Ending Capital	**$87,971**	$72,944	$65,026
Net Profit	**$37,971 (75.9%)**	$22,944 (43.9%)	$15,026 (32.1%)
Annualized Gain	70.9%	41.2%	30.2%
Total Commission	-$48,475	-$25,573	-$22,902
Exposure	209.7%	139.4%	103.0%

TRADE STATISTICS	ALL	LONG	SHORT
Number of Trades	24,238	12,787	11,451
Average Profit	$1.57	$1.72	$1.40
Avg Bars Held	27.9	28.9	~42
Winning Trades	8,387	4,434	3,953
Win Rate	34.6%	34.7%	34.5%
Avg Win	$48.12	$48.90	$47.24
Losing Trades	15,851	8,353	7,498
Loss Rate	65.4%	65.3%	65.5%
Avg Loss	-$23.06	-$23.33	-$22.77
Max Consec. Wins	17	22	35
Max Consec. Losses	27	84	41

RISK & QUALITY METRICS	ALL	LONG	SHORT
Max Drawdown	-$8,904 (-13.5%)	-$11,731	-$12,496
Wealth-Lab Score	29.3	23.5	23.1
Sharpe Ratio	2.05	1.46	0.98
Profit Factor	1.10	1.11	1.09
Recovery Factor	4.28	1.87	1.28
Payoff Ratio	1.88	1.89	1.87

Key insight: Despite a 65.4% loss rate, the strategy is profitable because the **payoff ratio of 1.88** means each winning trade earns nearly twice what a losing trade costs. The 2.05 Sharpe ratio and 4.28 recovery factor confirm the system meets professional quality thresholds (Sharpe ≥ 2.0; Yoshito Recovery ≥ 2.0).

TRADING ROBOTS

Advantages & Disadvantages

Advantages

+ **Speed.** Monitors hundreds of securities simultaneously, makes decisions and executes orders in milliseconds.

+ **Accuracy.** No button-press mistakes, no misread prices, no calculation errors.

+ **Inexhaustibility.** Works 24/7 without fatigue — never misses signals from exhaustion or loss of concentration.

+ **No Emotions.** Follows the algorithm precisely — never hesitates to cut losses, never gets greedy on winning streaks.

+ **Scalability.** Can simultaneously trade multiple strategies across hundreds of instruments.

Disadvantages

– **Cost.** Professional robots cost €1,000–5,000+ to develop. HFT may require colocation in exchange data centers.

– **System Failures.** If code has a bug, the robot repeats the same mistake systematically — multiplying losses.

– **Rigidity.** Cannot adapt to unprecedented market events outside its training parameters.

– **No Judgment.** Will drain capital following strategy signals during black swan events — requires human oversight.

– **Over-optimization Risk.** A curve-fitted strategy may excel in backtests but fail on future data.

THESIS ROADMAP

Seven Chapters

1 **Introduction**

Problem definition, 5 research questions, goals, motivation, and scope. Target: individual traders, hedge fund managers, banks, and financial institutions.

2 **Automated Trading on Stock Market**

Defines algorithmic trading. Quantifies market impact: 30–70% of Wall Street volume, 60% on Russian exchanges. Discusses quotation vs. market orders and exchange protective mechanisms against HFT overload.

3 **Trading Robots: Types & Platforms**

Covers automatic vs. semi-automatic robots (advisers, scanners, utilities). Evaluates platforms: QPILE, Excel+VBA, C++/Delphi, C#/S#+WealthLab. Presents 6 architecture options (A–F) from internet-connected broker to DMA colocation.

4 **Stock Market Analysis Tools**

Fundamental vs. technical analysis. Deep coverage of indicators: Moving Averages, Oscillators, RSI, Bollinger Bands, MACD, Fibonacci levels. Reviews MetaStock, Wealth-Lab, Omega as software facilitators.

5 **Trading Strategies Overview**

Classification: execution strategies (Iceberg, TWAP, VWAP) vs. speculative strategies (Trendline, Pullback, Breakout, Arbitrage, Scalping). Evaluates each for automation difficulty and speed requirements.

6 **ARP–Breakout Strategy Implementation**

The practical core: tool selection (QUIK + Wealth-Lab + C#/S#), architecture design, 14-parameter strategy description, Monte Carlo optimization, stock filtering (80 → 25), portfolio backtest over 2.5 years of NYSE data.

7 **Conclusions & Further Research**

Summarizes advantages/disadvantages of trading robots. Proposes future directions: multi-algorithm trading, strategy portfolios, strategy-of-strategies evaluation, automated market inefficiency detection.

PLATFORM SELECTION

Trading Robot Development Platforms

PLATFORM	SPEED	COMPLEXITY	BEST FOR	SUITABILITY FOR HFT
QPILE (QUIK)	~1 sec/iteration	Low	Simple robots, semi-automated utilities, learning	No
Excel + VBA	Moderate	Medium	Advanced investors, GUI-controlled robots	No
C++ / Delphi	Milliseconds	Very High	Professional HFT, scalpers, arbitrage robots	Yes
C# / S# + Wealth-Lab	>1M events/sec	High	Multi-core HFT, multi-exchange, professional systems	Yes
MetaStock / Omega	Moderate	Medium	Strategy backtesting, technical analysis, simple robots	No

Selection rationale: C# / S# (Stock Sharp) + Wealth-Lab was chosen for its native .NET performance, multi-core processing, emulation >1 million events/second, When-Do rule model, direct access to both Russian and international exchanges via Plaza II, and ability to run hundreds of concurrent algorithms.

FUTURE DIRECTIONS

Suggestions for Further Research

MULTI-ALGORITHM PARADIGM

This thesis used a mono-algorithm approach. Compare performance of single-strategy vs. multi-strategy automated systems. How do parallel strategies interact on the same portfolio?

STRATEGY PORTFOLIOS

Instead of a portfolio of stocks, create a portfolio of strategies — trade the same instrument with multiple complementary strategies simultaneously.

STRATEGY EVALUATION

Develop new dimensions for evaluating strategy performance beyond Sharpe ratio and recovery factor. How to measure strategy vitality over changing market regimes?

AUTOMATED INEFFICIENCY DETECTION

If markets are increasingly inefficient (Haugen 1995; Rosenberg & Reid 1998), can we automate the discovery of new market inefficiencies from real-time data? Strategy-to-generate-strategies.